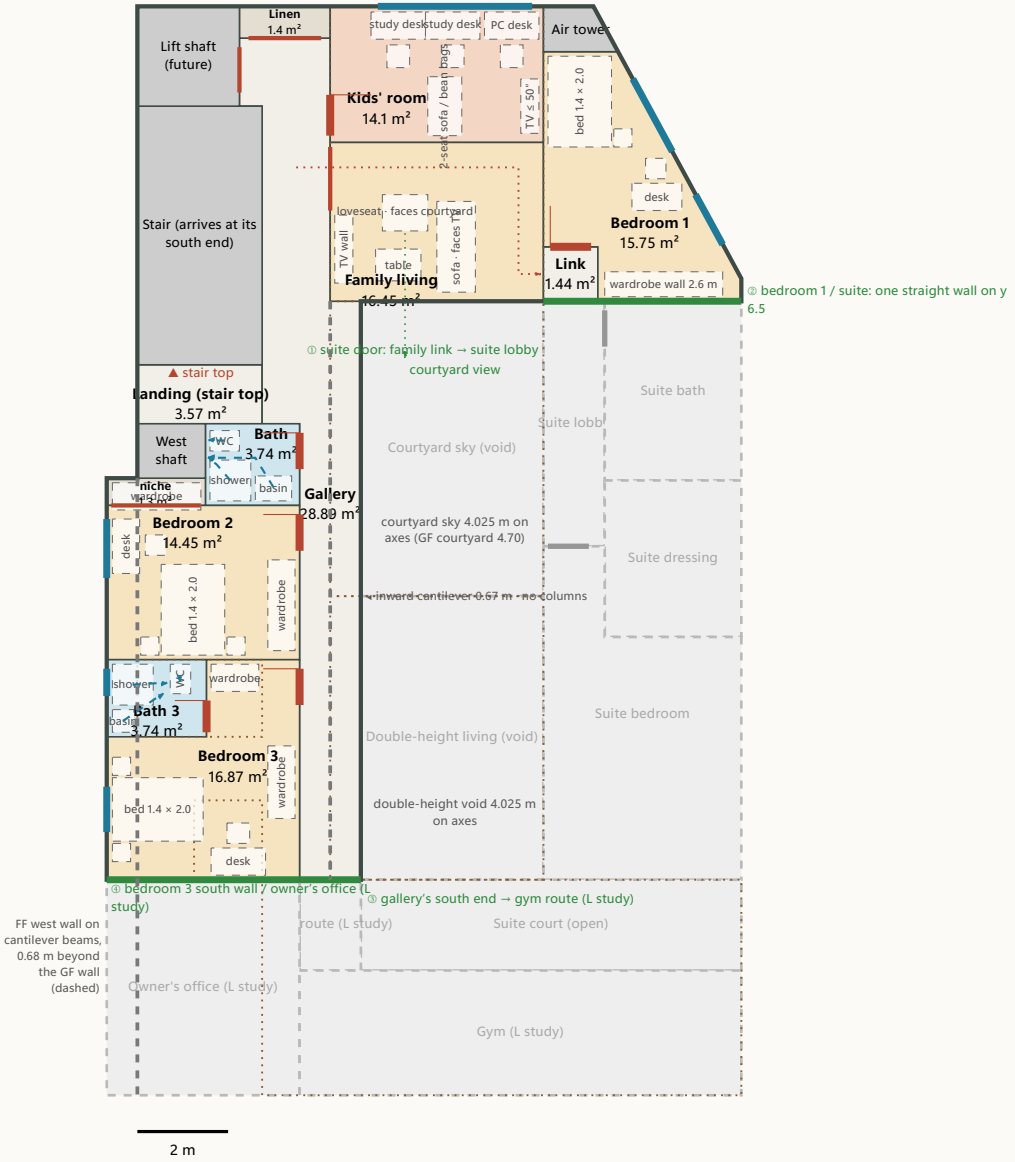
First floor · family part · focus study

Taif / Palm House · first floor V6.0 · local = main CSV coordinates (x east, y south) · wall centre lines · north up

First floor V6.0 · part 1 of 2: family part · split west projection, west rooms 4.25 m wide · part 2 (office, gym, suite court, master suite): studies/ff_suite.typ



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	2.75 × 1.3	3.57	3.14
Gallery (north part + on the projection)	polygon	28.89	25.92
Linen cupboard	2 × 0.7	1.4	–
Family bath (bed 1 + 2, family living)	2.08 × 1.8	3.74	2.96
Wardrobe niche (bed 2)	2.17 × 0.6	1.3	–
Bedroom 2	4.25 × 3.4	14.45	12.84
Bedroom 3 bath (en-suite)	2.2 × 1.7	3.74	2.92
Bedroom 3	polygon	16.87	14.97
Kids' room (study + gaming)	4.7 × 3	14.1	12.44
Family living	4.7 × 3.5	16.45	15.16
Family link	1.2 × 1.2	1.44	1.08
Bedroom 1	polygon	15.75	13.83
Fixed: stair, lift, west shaft, air tower		23.8	
Zone		**145.51**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, ≈ 50 mm balustrade on the void edges, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the routes · door swings clear · every room has a door reached from circulation · routes ≥ 1.0 m clear · suite door and gallery end on circulation

Key numbers

- **Split west projection:** 0.67 m **inward** over the courtyard and the double-height void (x 4.25–4.93, a plain cantilever, **no columns**) and 0.68 m **outward** beyond the GF west wall (x 0 → −0.68, y 10.4–24, the west wall on cantilever beams). Stair, lift, landing and west shaft rise from the ground floor; the west face steps out at y 10.4.
- **Courtyard sky** and **double-height void** 4.025 m on axes (the GF courtyard is 4.70); the ground-floor courtyard and living keep their full floor.
- **West rooms** 4.25 m on axes. **Gallery on the projection:** 1.35 m (½ wall + ½ railing off); wider at the stair top. **Circulation in the zone:** 33.9 m² (landing, gallery, family link); the walk through the living is not counted.
- **Family living:** 16.45 m², on the courtyard edge, no separate route. **Kids' room:** 14.1 m².
- **Bedrooms:** 2 · 14.45 m² · 3 · 16.87 m² + its own bath 3.74 m² · 1 · 15.75 m² (no bath). **Family bath** 3.74 m² for bedrooms 1 and 2, the family living and the kids' room.
- **Walked to the family bath door:** bedroom 2 ≈ 2.7 m · family living ≈ 7.7 m · bedroom 1 ≈ 16.2 m. Bedroom 3 → its own bath ≈ 2.7 m.

Plan: red = doors (thin = opening without a leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below (the full courtyard and living floors) · grey dashed = the two GF wall lines the cantilevers start from (x 4.25 courtyard edge, x 0 west wall) · grey = neighbours · dashed boxes = furniture and fixtures.

Family part · the west projection and the rooms

The split west projection

- **Where:** 0.67 m inward along the west edge of the courtyard and the double-height living (x 4.25–4.93, y 6.5–19.25) and 0.68 m outward beyond the GF west wall (x 0 → −0.68, y 10.4–24). The west band (bedrooms 2 and 3, bedroom 3 bath, wardrobe niche, gallery, and in the L the office and the gym route) stands on it; stair, lift, landing and west shaft rise from the ground floor, so the west face steps out at y 10.4, just south of the shaft.
- **Widths:** rooms 4.25 m on axes (≥ 4.0 m clear), gallery 1.35 m (≥ 1.0 m clear). The site limits the outward part (the GF outer face to the M line); the even split leaves the gap shown on design_ff.pdf.
- **Inward part:** a plain cantilever from the courtyard edge x 4.25 (back span to the GF riwaq wall x 2.75), carrying only the gallery and its railing: **no edge beam, no columns**.
- **Outward part:** the FF west wall on **cantilever beams** beyond the GF west wall, y 10.4–24. It shades the GF west wall from the afternoon sun and roofs the guest entrance (foyer door).
- **Gallery:** a railing on the void edge, wider at the stair top; every west room door opens from it; no room opens onto the stair landing.

Kids' room · family living · bedroom 1

- **Kids' room** 14.1 m^2: two study desks + PC desk under the north window, a gaming corner (TV on the east wall), solid door from the gallery.
- **Family living** 16.45 m^2: a view, not a path; the walk to the family link passes behind and beside the sofa.
- **Bedroom 1:** one room, no bath, its south wall straight on y 6.5; it uses the family bath.

Family bath · 2.075 × 1.8 m, beside the west shaft

- WC on the shaft wall, shower, basin by the door; door from the gallery (opening in, swing clear), not onto the landing. For bedrooms 1 and 2, the family living and the kids' room. Its west wall is the shaft (x 1.5), its east wall the band's gallery wall.

Bedroom 2 · 4.25 × 3.4 m + wardrobe niche 2.175 m

- **Bed 1.4 × 2.0** with its head on the south wall and bedside tables on both sides; **wardrobes** in the niche beside the shaft and on the east wall; **desk** under the west window. Door from the gallery (swing clear), ≈ 2.7 m from the family bath.

Bedroom 3 · 16.87 m^2 + its own bath 3.74 m^2

- An entry with a wardrobe, then the room: **bed 1.4 × 2.0** head on the west wall with two bedside tables, wardrobe on the east wall, desk on the south wall. Door from the gallery (swing clear).
- **Own bath (en-suite)** off its entry: shower, WC, basin; ≈ 2.7 m from the bedroom door. It stands **over the GF family WC** (walls on the GF lines y 14.4 and 16.1); its west part is on the outward cantilever, the drains run in the floor to the WC stack.

Gym route and the owner's office → the L study

- The gallery ends at y 19.25; from there the **gym route** and the **owner's office** belong to the L study (studies/ff_suite.typ), with the gym, the court and the suite.

Interfaces with the L study (fixed)

- ① suite door at the end of the family link (y 6.5); ② bedroom 1 / suite: one straight wall on y 6.5; ③ the gallery's south end (y 19.25), where the gym route starts; ④ bedroom 3's south wall (y 19.25), a wall to the owner's office (no window).

Family part · routes, walls, drains, open items

Routes (≥ 1.0 m clear)

Route	Axes	≈ clear	Deducted
Landing	1.3	1.2	½ wall on the south side; the north side is the stair arrival
Gallery north (beside the stair)	1.5	1.3	½ wall each side
Gallery on the projection	1.35	1.22	½ wall (rooms side) + ½ railing (void side)
Walk behind the sofa (in the living)	1.1	1	½ wall on the kids' room side; the sofa side is open
Walk beside the sofa to the family link	1.1	1	½ wall on the bedroom 1 side; the sofa side is open
Family link	1.2	1	½ wall each side (bedroom 1 to the north and east)

Walking distance	m
Stair top → kids' room door	8.8
Stair top → family living (opening)	8.7
Stair top → bedroom 1 door	15.9
Stair top → ⓘ suite door	15.9
Stair top → family bath door	5
Stair top → bedroom 2 door	6.8
Stair top → bedroom 3 door	10.2
Stair top → gallery's south end (L)	13.8
Bedroom 2 door → family bath door	2.7
Family living → family bath door	7.7
Bedroom 3 door → its own bath door	2.7
Bedroom 1 door → family bath door	16.2

Stair top → every room, the suite door and the gallery's south end runs only on the landing, the gallery, the family link and the walking line through the living (checked: every door's approach is on circulation, no furniture in a route).

Engineer

- **Inward cantilever:** 0.67 m slab beyond the courtyard edge x 4.25, y 6.5–19.25, back span to the GF riwaq wall x 2.75; it carries only the gallery and its railing. **No columns, no edge beam** expected; the x 4.25 beam line carries the rooms' former wall line and the cantilever. To confirm: slab depth, the x 4.25 beam and the railing fixing.
- **Outward cantilever:** the FF west wall stands 0.68 m beyond the GF west wall, y 10.4–24: **wall on cantilever beams** (one at each cross wall, back span to x 2.75, an edge beam under the wall). To size: beam depth, the wall's line load, tip deflection, the south-west corner.

Walls off the ground-floor lines

Ground-floor lines here: x 0, 1.25 (guest WC), 1.5 (shaft), 2.25, 2.75, 4.25, 8.95 · y 2.2, 7.9, 10.4, 12.9, 14.4, 16.1, 17.5, 19.25, 21.4 (foyer) (west band); the women's majlis (x 4.25–8.95, y 0–6.5) and the men's majlis (x 2.75–13.311, y 19.25–24) have no inner wall; east part (bedroom 1, family link): over the GF maid wing, laundry and the kitchen's north wall.

- **Off:** y 3.0 (kids' room / living, over the women's majlis); y 11.0 (family bath and niche / bedroom 2) and x 1.5 south of the shaft (niche); **x 3.575** (the rooms' east wall, on the slab between x 2.75 and 4.25); **x −0.675** (the west wall, on the cantilever beams); x 1.525 (bedroom 3 bath / entry).
- **On:** y 14.4 and 16.1 (bedroom 3 bath = the GF family WC walls), x 1.5 / y 10.4 (shaft), y 6.5 (bedroom 1 / suite), the outer walls north of y 10.4.
- The GF riwaq wall x 2.75 carries the floor; the rooms span over the riwaq.

Drains (blue dashed)

- **Family bath:** WC on the west shaft wall, shower and basin in the bath floor to the shaft; floor over the GF utility room and hospitality kitchen, nothing in their ceilings.
- **Bedroom 3 bath:** over the GF family WC: WC straight into its stack, shower and basin on the outward cantilever, drains in the floor build-up. Wet room on a wet room.
- **Bedroom 1:** no bath. No east duct (the suite bath drains to a stack outside the east wall, see ff_suite.typ).

Open items (all in docs/OPEN_ITEMS.md)

- **Before construction drawings:** municipality / engineering office to confirm the first-floor projection this close to the M line (gap on design_ff.pdf); if not allowed, fall back to the full inward projection on three columns at x 5.6 (tag v5.0).
- **Engineer:** the wall on cantilever beams; the inward cantilever without columns; walls off the GF lines.
- **Built-in cupboards in niches** (linen cupboard, bedroom 2 wardrobe niche): joinery boxes scribed to the walls; decide with the carpenter.
- **Courtyard light:** a daylight study of the courtyard and the double-height living.